Particle Filter Robot Localization

Intelligent Robotics, Assignment 1 · Student ID 2929562 · Code: github.com/worshub05/My_localization

1. Problem Statement

Estimate the pose (x, y, θ) of a differential-drive robot driving a 6 m × 4 m loop (20.5 m, 66 s) in Gazebo, using only noisy wheel odometry and noisy range-bearing readings of 8 known landmarks. Odometry alone drifts without bound, so both sources are fused.

2. Methods

Setup. ROS 2 Humble and Gazebo Fortress via `ros_gz_bridge`; a driver node commands 0.4 m/s straights and 0.5 rad/s turns. Ground truth is used only for evaluation.

Uncertainty. Each odometry step gets a systematic bias (+2% distance, +4% rotation) plus Gaussian noise proportional to the motion. A 5 Hz sensor returns the ID, range and bearing of landmarks within 6 m (σ = 0.10 m, 0.05 rad); IDs and the map are known.

Particle filter (500 particles). *Predict:* move each particle by the odometry increment plus noise, inflated to absorb the bias. *Update:* weight by the Gaussian likelihood of measured vs expected range and bearing. *Resample:* low-variance, when the effective sample size drops below N/2. *Estimate:* weighted mean pose.

3. Results

The filter cut position RMSE from 0.466 m to 0.061 m (7.6×) over the 66.4 s drive and ended 0.033 m from the true pose, versus 0.851 m for odometry alone.

Method	Pos. RMSE (m)	Pos. max (m)	Pos. final (m)	Heading RMSE (°)	Heading final (°)
Dead reckoning	0.466	0.860	0.851	6.33	10.30
Particle filter	0.061	0.122	0.033	0.98	0.15

Analysis. Dead-reckoning heading error jumps about 3° at each 90° turn (mainly the 4% rotation bias), so position error grows to 0.86 m. The filter stays below 0.12 m, peaking briefly at turns. Sensor noise, inflated filter noise and few visible landmarks at the corners limit precision.